EVID: Reference-free Evaluation of Action Fidelity in Videos from Robot World Models via Inverse Dynamics

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Abstract

Action-conditioned video world models predict the video a robot would record when it executes a given sequence of actions, and they are compared mostly with frame-level appearance metrics such as PSNR, SSIM, and LPIPS against the recorded video. These metrics leave two gaps: they need the real video, which does not exist for a new action sequence, and they score how the frames look, not whether the robot in the video performs the commanded motion or whether a real arm could perform it. We reproduce IRASim and its baselines and add Cosmos-Predict2.5 with and without action input on 105 Bridge and 57 Language-Table clips. In a controlled test where videos are generated from deliberately corrupted actions, SSIM rates a static copy of the first frame above every corrupted video and as high as the video from the correct actions, and PSNR and LPIPS rate it above the three strongest corruptions, whereas an end-effector track compared with the real video degrades steadily as the corruption grows. We propose EVID, which closes the loop from actions to video and back to actions: the Cosmos3-Nano inverse dynamics model reads the end-effector trajectory from the generated video, and we measure how far it deviates from the commanded trajectory and check, with the inverse kinematics of the WidowX 250 arm, whether a real robot could execute it. Without any real video, EVID separates real videos from a static one by a factor of six, grows monotonically with the action corruption, and ranks the models in the same order as the reference-based metrics. It shows that Cosmos-Predict2.5 conditioned on actions follows them most faithfully, that the same backbone without actions moves the robot no better than a static video, and that almost every arm motion in the generated videos is executable by the real arm; the most frequent errors are instead objects that change shape or vanish, which neither EVID nor the appearance metrics detect. Code, videos, and an interactive results viewer are available at hoangducanh1865.github.io/PI.

1. Introduction

A video world model for robot manipulation takes the first camera frame I_0 and a sequence of actions $a_{1:T}$ and predicts the video $\hat{V}$ the robot would record. If the prediction is right, the model can replace a physics simulator for policy evaluation, planning, and data generation [10, 18, 20]. Such models are compared mostly with frame-level appearance metrics against the recorded video, namely PSNR, SSIM [17], and LPIPS [19], or with FVD [15] over a set of videos. These metrics leave two gaps. First, they need the real video of the same actions, which does not exist when the model generates the outcome of a new action sequence, the very use of a simulator. Second, they score how the frames look, not whether the robot in the video performs the commanded motion, whether a real arm could perform it, or whether objects move only when touched.

We reproduce IRASim [20] on Bridge [16] and generate 105 clips again from deliberately corrupted actions: SSIM rates a static copy of the first frame above every corrupted video and as high as the video from the correct actions, and PSNR and LPIPS rate it above the three strongest corruptions, whereas the end-effector track in the image, compared with the track in the real video, degrades steadily as the corruption grows. Comparing tracks, however, needs the real video, which a simulator does not have for a new action sequence. What it does have is the commanded actions, so a generated video can be checked against them instead of against a real video, which is the idea of our metric. What it does have is the commanded actions, so a generated video can be checked against them instead of against a real video, which is the idea of our metric.": What is the purpose of this claim? How does this relate to the above claims?

Our main contributions are: (1) an evaluation framework that lists, for each metric group, what it measures and what it cannot measure, with a perturbation test that checks whether a metric notices wrong actions, and (2) EVID, a reference-free metric that closes the loop from actions to video and back: an off-the-shelf inverse dynamics model reads the end-effector trajectory from the generated video, its output is converted to end-effector poses with the same formulas its authors use to prepare their training actions, so nothing is fitted, and EVID measures the deviation from the commanded

trajectory and checks whether the WidowX 250 arm could reach every pose within its joint and speed limits.

2. Related work

UniSim [18] learns a video simulator of real-world interaction from mixed internet, simulation, and robot data, generating the visual outcome of language instructions or low-level controls, and trains policies inside it that transfer to real robots. IRASim [20] generates robot videos from a first frame and a trajectory of end-effector actions with a diffusion transformer [14] that conditions each frame on its own action. It is also used to evaluate a policy, by generating videos of the policy's rollouts and letting humans label each one as a success or a failure, and to plan, by generating the outcome of several trajectories sampled from a policy and executing the one a reward model scores highest. Cosmos [10] provides pretrained video world foundation models for physical AI, and its Cosmos-Predict2.5 release includes a model post-trained on Bridge for action-conditioned generation. None of these checks, without a real video, whether the robot in a generated video performs the commanded actions.

Inverse dynamics models, which predict the action taken between two consecutive frames, are the usual way to turn videos into actions. VPT [1] trains such a model on a small labelled set to label large unlabelled video collections, UniPi [3] applies it to text-conditioned video plans to obtain robot actions, and DreamGen [5] uses it to label videos generated by a world model as training data for visuomotor policies. These works trust the actions read from the videos; we compare them with the commanded ones to evaluate the video.

VideoPhy [2] rates whether text-to-video generations follow physical commonsense with human judges and an automatic judge. Physics-IQ [9] gives a video model the first seconds of a real physical experiment, such as falling dominoes, compares where and how much motion appears in its continuation with the recorded one, and finds that visual realism does not imply physical understanding. Like our reference-based metrics, it needs the real video, and neither benchmark involves robot actions. SIMPLER [7] evaluates real-robot policies in physics simulators matched to real setups, including the WidowX setup of Bridge; it is the kind of hand-built simulator that video world models aim to replace, and a place where motions read from generated videos could be replayed with physics. Table 5 compares these sources by inputs, outputs, evaluation methods, assumptions, and limitations.

Open questions. The sources leave three questions open: whether a frame that looks real implies that the robot followed the commanded actions, whether objects obey contact by moving only when touched and keeping their identity, and whether the robot motion in the video can be executed by the real arm. We measure the first and the third with EVID, and the second only through vision-language judges.

3. Reproducing IRASim

Pipeline. IRASim [20] predicts 16 frames from the first frame and 15 actions. Each action is the end-effector motion between two frames, expressed in the end-effector frame of the earlier one: the position change $R_{t-1}^{\top}(p_t - p_{t-1})$, the Euler angles of the rotation change $R_{t-1}^{\top}R_t$, and the gripper state, with the arm part scaled by 20. The SDXL VAE encodes each 256×320 frame into a $4\times32\times40$ latent, and a diffusion transformer with patch size 2, IRASim-XL/2 [14], denoises the latents of frames 1 to 15 while the clean latent of frame 0 serves as the condition. Actions enter through adaptive layer norm with zero-initialised gates: Frame-Ada modulates the tokens of each frame by that frame's action, while Video-Ada modulates every frame by one embedding of the whole sequence. Training minimises the mean squared error between the added and the predicted noise on frames 1 to 15, with a linear schedule of 1,000 steps, 300k iterations, and learning rate 10^{-4}, and sampling uses PNDM with 50 steps. The baselines of [20] are LVDM, a latent video diffusion model with a UNet, and VDM, a diffusion model in pixel space, both conditioned on the whole action sequence.

Setup. We use the code, checkpoints, and evaluation settings of [20]. Clips are 16 consecutive frames at 5 Hz, about 3 s, from Bridge [16] test episodes, taken as the first episodes of each instruction verb: 105 clips over 12 verbs. Three costly experiments, namely six seeds for best-of-N selection, the model without actions, and the judges, use a fixed subset of 24 of these clips over 8 verbs, the core clips. On Language-Table [8] we use 57 clips of 288×512 frames from 48 episodes. IRASim models run on Kaggle T4 GPUs, and the Cosmos models and the inverse dynamics model on one A100 GPU.

Reproduction details. We score a subset of the test episodes rather than the full test set. We do not compute FVD [15]: it compares the feature distributions of two sets of videos, so it gives no score per clip and no paired comparison, its estimate is unreliable for about a hundred videos, and it does not depend on which actions were commanded, the question we study. In the IRASim code the initial noise of a clip depends on the other clips in its batch, so the same seed gives different samples once the clip list changes; we treat each seed as one sample and report Frame-Ada as the mean of two seeds. Our PSNR is lower than reported for every model while the ranking of the models is unchanged (Tab. 3). The evaluation code of [20] scores the same frames, 1 to 15, against the same unprocessed real videos, so the off-

Model	Conditioning on frame 0 and	Output
IRASim [20] and its baselines, 50 sampling steps		
Frame-Ada	15 actions, one per frame	16 frames, 256×320
Video-Ada	15 actions, one embedding	16 frames, 256×320
LVDM	15 actions, one embedding	16 frames, 256×320
VDM	15 actions, one embedding	16 frames, 256×320
Cosmos-Predict2.5 [10], 35 sampling steps		
With actions	chunks of 12 actions	13 frames per chunk, 256×320
Without actions	instruction text	49 frames at 16 fps, 480×640

Table 1. Models compared on Bridge. Cosmos with actions generates two chunks with guidance 0, of which frames 0 to 15 are kept; without actions it uses guidance 7 and runs on the 24 core clips only.

set is not a difference of protocol. It comes mostly from the clips: on 64 clips from 40 test episodes drawn at random, Frame-Ada reaches 24.85 dB PSNR and 0.828 SSIM over two seeds, within 0.43 dB and 0.005 of the reported values, so the first episodes of each verb that we score are harder than the average test clip.

4. Evaluation framework

Models and references. Table 1 lists the inputs and outputs of the models we compare, and Fig. 1 shows how each video is scored. Cosmos-Predict2.5 [10] enters in two versions of the same 2B backbone: the model post-trained on Bridge with actions, and the base model given the first frame and the instruction text but no actions, which tests whether the action input matters. Two references bound every score: the real video of the same actions is the ceiling, and the first frame held still for 16 frames is the floor, with perfect image quality and no motion. Cosmos videos are resampled to the IRASim format of 16 frames at 256×320; for the base model we keep every third of its 49 frames at 16 fps, which spans the same 3 s.

Metric groups. Table 2 lists what each group measures and what it cannot measure. Every metric is computed per clip, and two models are compared with a paired Wilcoxon signed-rank test over the same clips; we call a difference significant when $p < 0.05$, that is, when a difference this consistent across clips would arise by chance less than 5% of the time if the two models were equally good. The 2-D track score follows the end-effector in the image. CoTracker3 [6] tracks a 20×20 grid of points from frame 0 through the real and the generated video, and the end-effector point is the moving point whose real track is best explained by the recorded end-effector position through an affine camera; two vision-language models confirm that the point lies on the gripper. NDTW [4] compares the generated and the real track of that point,

$$\mathrm{NDTW} = \exp\left(-\frac{\mathrm{DTW}(\hat{x}_{0:15}, x_{0:15})}{16\, d_{\mathrm{th}}}\right), \qquad (1)$$

where DTW sums the pixel distances between the two tracks after aligning them in time, and $d_{\mathrm{th}} = 20$ px; NDTW is 1 for identical paths and tends to 0 for unrelated ones.

5. EVID: reading the actions back

Idea. Every metric in Tab. 2 except the judges needs the real video of the same actions, and none checks whether the motion shown could be performed by the robot. EVID uses only what a simulator always has, the commanded actions: an inverse dynamics model reads the end-effector trajectory back from the generated video, and we measure how far it deviates from the commanded trajectory and whether the arm's inverse kinematics can execute it (Fig. 1). A video that shows the commanded motion closes the cycle from actions to video and back with a small error. We use NVIDIA Cosmos3-Nano [11] in its inverse dynamics mode for the Bridge robot, without any training of ours.

From the model output to a trajectory. Let T_t be the pose of the end-effector at frame t, a 4×4 matrix holding its rotation and position. For each pair of consecutive frames, the model returns 10 numbers $\hat{a}_t$ that encode the motion between them, $\Delta_t = T_{t-1}^{-1} T_t$: three for the translation, six for the rotation, and one for the gripper command. NVIDIA's encoder [12] scales each number a to about $[-1, 1]$ with the 1st and 99th percentiles q_{01} and q_{99} it publishes for this robot,

$$\hat{a} = 2\,\frac{a - q_{01}}{q_{99} - q_{01}} - 1. \qquad (2)$$

We solve Eq. (2) for a to recover each motion $\hat{\Delta}_t$, then add the motions up from the true first pose,

$$\hat{T}_0 = T_0, \qquad \hat{T}_t = \hat{T}_{t-1}\,\hat{\Delta}_t, \quad t = 1, \ldots, 15, \qquad (3)$$

and the positions of $\hat{T}_{1:15}$ form the trajectory $\hat{p}_{1:15}$ read from the video. Nothing is fitted: these are NVIDIA's own formulas run backwards, and passing recorded Bridge states through their encoder and our decoder returns them within $2 \cdot 10^{-4}$ mm. The only adjustment is a convention: the Bridge states used by IRASim measure the roll angle of the gripper with an offset of π from the data NVIDIA trained on.

Cycle error. The cycle error of a clip is the mean distance between the end-effector positions read from the video and the commanded ones,

$$E_{\mathrm{cycle}} = \frac{1}{15} \sum_{t=1}^{15} \|\hat{p}_t - p_t\|_2, \qquad (4)$$

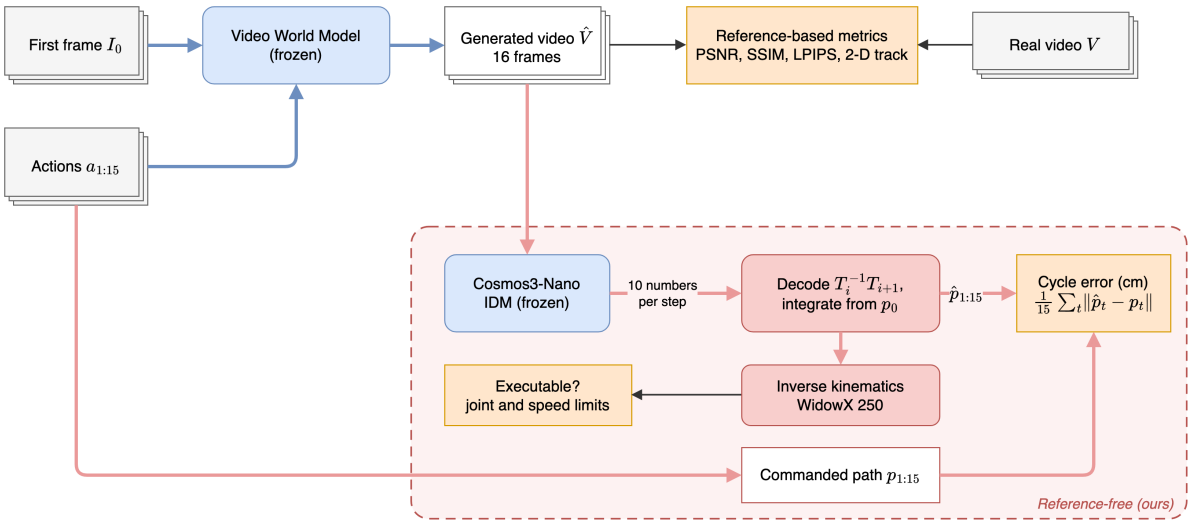


Figure 1. Evaluation pipeline. Both networks are frozen; reference-based metrics need the real video, while EVID (dashed box) needs only the commanded actions: it measures the deviation of the trajectory read back from the video and checks whether the WidowX 250 arm could execute it.

Group and metrics	Measures	Cannot measure
Visual quality: PSNR, SSIM [17], LPIPS [19] against the real clip	how close each frame looks to the recorded frame	whether the robot moved as commanded, since a still video keeps a high SSIM and small shifts are punished like large errors
Temporal consistency: DINOv2 [13] similarity of adjacent frames and to frame 0, optical-flow warp error	smoothness and identity stability across frames	correct motion, since the still video scores perfectly
2-D track: CoTracker3 [6] end-effector track against the real clip, NDTW [4], pixel error	whether the gripper moves along the same image path as in the real clip	depth and orientation; it needs the real clip and an end-effector point checked by vision-language models
Physical plausibility and outcome: two vision-language judges	violations a judge sees, such as objects that vanish or change and a robot that deforms, and whether the final state matches the real clip	a reliable verdict, since the judges differ in strictness and change answers between runs
Action cycle consistency: EVID, ours	whether the 3-D end-effector trajectory read from the video matches the commanded one, in cm, and whether the arm could execute it, without a real video	object motion and contact, collisions and torques; it is bounded by the error of the inverse dynamics model, 1.50 cm on real clips

Table 2. Metric groups. Task completion is covered by the outcome question of the judges.

in centimetres, lower being better.

Executability. For each of the 16 poses read from the video, inverse kinematics searches the six joint angles of the WidowX 250, the 6-DoF arm of Bridge, that place the gripper at that pose. A clip is executable when every pose has a solution, every solution lies within the joint limits, and no joint has to turn faster than its velocity limit of π rad/s between two poses 0.2 s apart, one step of the 5 Hz Bridge control. The kinematic model, the solver with its tolerances of 1 mrad and 1 mm, its three starting guesses, and the limits are those of the arm's Interbotix driver. That solver rejects 4 of the 105 trajectories recorded on the real arm, all low and steeply pitched, so we add three fallbacks with the same tolerances and limits: 125 more starting guesses, the solution closest to the previous frame when a step looks too fast, and a search over the solutions of the first pose when a clip still fails. Without the roll convention above, the check rejects 381 of the 384 recorded poses of the 24 core clips, so it is sensitive to errors in the pose frame.

Use for selection. Among the metrics of Tab. 2, only EVID, the judges, and the temporal scores work without a real video, and only EVID checks the commanded actions, so it can rank several samples generated from the same actions by their cycle error and keep the best one, which we test as best-of-N selection.

Model	Visual			2-D track		EVID	Reported
	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	px↓	cm↓	PSNR↑
Bridge, 105 clips							
IRASim-Frame-Ada	23.87	0.808	0.083	0.709	9.9	1.82	**25.28**
IRASim-Video-Ada	23.52	0.804	0.087	0.682	10.5	1.95	24.73
LVDM	22.62	0.788	0.103	0.593	14.9	2.54	23.55
Cosmos-Predict2.5, actions	**25.74**	**0.859**	**0.063**	**0.754**	**7.4**	**1.64**	-
Real video	-	-	-	1	0	1.50	-
Still first frame	19.80	0.805	0.141	0.156	58.7	9.06	-
Language-Table, 57 clips							
IRASim-Frame-Ada	**27.15**	0.861	**0.059**	**0.702**	**9.2**	-	**28.82**
IRASim-Video-Ada	23.04	0.833	0.104	0.304	37.2	-	23.89
LVDM	26.44	**0.863**	0.066	0.646	10.9	-	28.25
Still first frame	22.35	0.879	0.103	0.131	80.7	-	-

Table 3. Main results. Visual metrics over frames 1 to 15; 2-D track on the clips with an end-effector point, 96 on Bridge; EVID: cycle error, not available on Language-Table. Frame-Ada: mean of two seeds, whose standard deviation is at most 0.17 dB PSNR and 0.03 cm. Reported: PSNR in [20] on its test set. Best in bold, second underlined, references in italics excluded.

6. Results

Bridge, 105 clips. Among the IRASim models, Frame-Ada leads, Video-Ada is second, and LVDM third on every metric (Tab. 3); the differences are significant on PSNR, 2-D NDTW, and EVID, and the two seeds of Frame-Ada do not differ significantly from each other. Cosmos-Predict2.5 with actions has the best mean on every metric and a significantly higher PSNR than Frame-Ada, 1.9 dB; its leads on NDTW and EVID, 0.045 and 0.18 cm, are small and not significant against both seeds. Its EVID error of 1.64 cm is within 0.14 cm of the error on the real videos, the floor set by the inverse dynamics model. SSIM separates no model from the still video.

Language-Table, 57 clips. On table-top pushing in 2-D (Tab. 3, bottom), Frame-Ada again leads LVDM, significantly on PSNR and less clearly on NDTW, and SSIM ranks the still frame first, 0.879 against 0.861 to 0.863, because most of the image is a static table. Video-Ada is significantly below LVDM on every metric and close to the still floor on motion. The PSNR reported by [20] shows the same gap, which indicates a property of the model rather than of our reproduction. EVID is not available here, because the inverse dynamics model has no Language-Table domain, and Cosmos-Predict2.5 provides no action-conditioned model for Language-Table.

Perturbation test. To check whether a metric notices wrong actions, we regenerate the 105 clips with Frame-Ada from corrupted actions, namely random walks of 0.5 and 1.5 cm per frame on the position, mirrored motion, and all actions set to zero, and score them against the real clips of the original actions (Tab. 4). NDTW and EVID both degrade steadily as the corruption grows, place zero actions

Actions given	Visual			2-D	EVID	
	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	cm↓	Exec.↑
IRASim-Frame-Ada, seed 3407						
Original	**23.95**	**0.809**	**0.082**	**0.699**	**1.81**	99.0
Noise 0.5 cm	21.65	0.782	0.107	0.557	2.57	99.0
Noise 1.5 cm	18.93	0.744	0.152	0.256	5.46	99.0
All zero	19.01	0.748	0.160	0.170	8.88	100
Mirrored	18.22	0.730	0.179	0.093	14.56	93.3
Reference						
Still first frame	19.80	0.805	0.141	0.156	9.06	100

Table 4. Perturbation test on the 105 Bridge clips, scored against the real clip of the original actions; NDTW on the 96 clips with an end-effector point; Exec.: percentage of clips whose motion read from the video is executable by the WidowX 250, not ranked since it is a necessary condition rather than a quality score. Best in bold, second underlined, reference excluded.

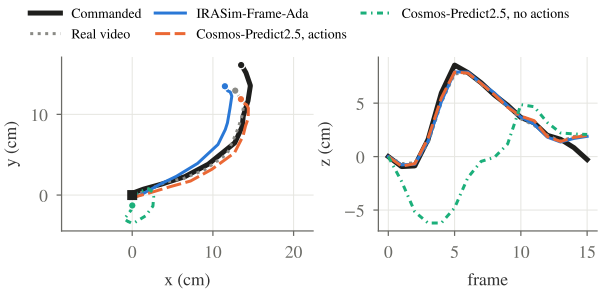


Figure 2. EVID on the clip "take banana out of colander": end-effector path read back from each video against the commanded path, from above (left) and as height over time (right). EVID errors: real video 1.63 cm, Frame-Ada 1.86, Cosmos-Predict2.5 with actions 1.82, without actions 15.53.

at the level of the static video, 0.170 against 0.156 and 8.88 against 9.06 cm, and place mirrored motion beyond it; EVID does so without the real video. The appearance metrics do not: SSIM rates the static video significantly above every corrupted generation and as high as the generation from the original actions, and PSNR and LPIPS rate it significantly above the three strongest corruptions, so a higher appearance score does not mean that the robot followed the actions.

Validating EVID. We check in four ways that EVID measures what it should. On real videos its error stays at the floor set by the inverse dynamics model, 1.50 cm for clips that move 25 cm on average, while a still video scores six times worse, as poorly as assuming no motion at all, so the model reads the motion from the frames rather than from a prior. It grows steadily as the actions are corrupted (Tab. 4). It ranks the models in the same order as the reference-based scores, each step of the ranking being significant (Tab. 3). Across generated clips it correlates with LPIPS, but its ex-

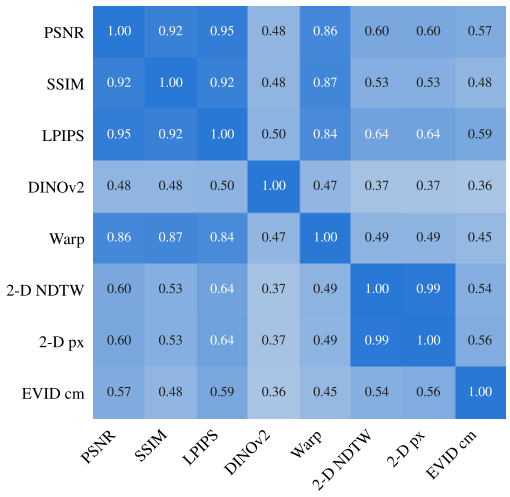


Figure 3. Spearman correlation between metrics over the 502 generated Bridge clips scored by every metric, each oriented so that higher is better.

cess over the error on the real video of the same clip barely does, so it measures something the appearance scores do not. Figure 2 shows one clip: the paths read from the real video and from both action-conditioned models stay within about 2 cm of the commanded one, while the model without actions moves the arm elsewhere.

Metric agreement. Over the 502 generated clips scored by every metric (Fig. 3), the appearance metrics PSNR, SSIM, LPIPS, and the warp error form one tight group, with Spearman correlations of 0.84 to 0.95, which the trajectory metrics join only moderately, at 0.45 to 0.64. EVID, computed without the real video, correlates as much with the 2-D track, 0.54 and 0.56, as with the appearance metrics, 0.48 to 0.59. All correlations are positive partly because a hard clip is hard for every metric; the perturbation test (Tab. 4) shows where the two groups disagree.

Executability. The executability check of EVID asks only whether the WidowX 250 could perform the arm motion read from a video, not whether that motion is the commanded one, so a still video passes trivially. It accepts 104 of the 105 trajectories recorded on the real arm. On generated videos it accepts 99% of clips for every action-conditioned model, the one failure being a clip that the inverse dynamics model misreads on the real video as well, and it rejects 7% of the videos generated from mirrored actions (Tab. 4). The arm motion in these videos is therefore almost always feasible; their failures lie in following the actions, which the EVID error measures, and in objects, which none of our metrics measures reliably.

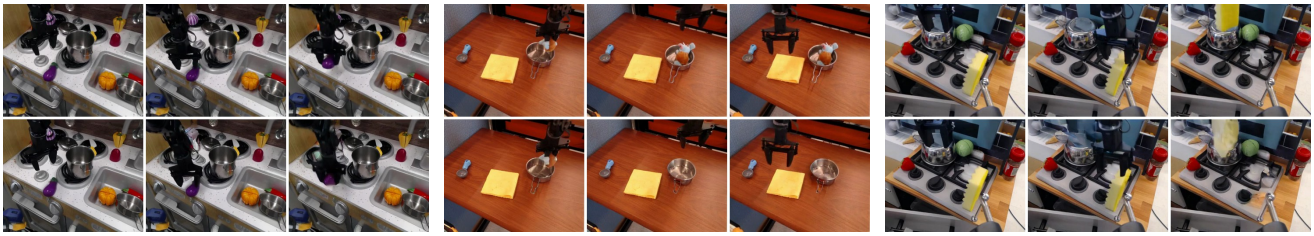
Action input matters. On the 24 core clips, Cosmos-Predict2.5 base, given the first frame and the instruction but no actions, produces smooth videos, with a DINOv2 adjacent-frame similarity of 0.916 against 0.952 for real clips, whose robot does not follow the recorded motion: its EVID error of 10.30 cm does not differ significantly from the still video, 9.52 cm on these clips, while the same backbone with actions reaches 1.74 cm. Its PSNR of 13.5 dB is also far below every action-conditioned model, partly because of resampling its 49 frames at 16 fps to our format. On the clip "move the toy and put it in the vessel" it even replaces the WidowX with a different, white robot arm, a failure of embodiment that the commanded actions prevent.

Temporal consistency and judges. On the 105 clips, DINOv2 adjacent-frame similarity is 0.952 for real clips and 0.925 to 0.935 for the action-conditioned models, while the still video scores 1.000, so these metrics measure stability rather than correct motion. On the 24 core clips, two vision-language judges, gemini-3.8-flash and gpt-6.1-sol, rate physical plausibility and whether the final state matches the real clip; they agree on the ranking but not on strictness, flagging 21% and 4% of real clips. The most frequent violation for every latent model is an object that changes or vanishes, in 46% and 17% of Frame-Ada clips, and the still video is judged plausible every time, caught only by the outcome question with 0.00 and 0.04.

Best-of-N selection. For each of the 24 core clips we generate six Frame-Ada videos and keep the one with the lowest EVID error. On the independent reference-based scores the selection brings no significant gain: PSNR is 24.02 against 23.93 dB for the average seed and NDTW 0.717 against 0.709, while an oracle choosing by PSNR itself, which needs the real video, would reach 24.86 dB. Seeds of Frame-Ada differ little, so there is little to select from; the selection may matter for models whose samples differ more.

7. Strengths and limitations of IRASim

Strengths. *S1, the robot follows the actions.* The gripper of Frame-Ada follows the real image path, with NDTW 0.709 and a mean error of 9.9 px over 96 clips, and the 3-D path read back from its videos is within 1.82 cm of the commanded one, against 1.50 cm for the real videos. *S2, per-frame conditioning helps.* With the same backbone and data, modulating each frame by its own action in Frame-Ada beats one embedding of the whole sequence in Video-Ada on PSNR and 2-D tracking over 105 paired clips. *S3, stable and reproducible.* Temporal consistency is close to the real videos, two seeds give statistically equal scores, and a rerun with the same batches reproduces the videos exactly.



(a) "put eggplant in pot": success
PSNR 23.33, EVID 1.18 vs 1.00 cm

(b) "put the toy in the vessel": the toy vanishes
PSNR 24.55, EVID 2.57 vs 2.52 cm

(c) "wipe pot with sponge": the gripper dissolves
PSNR 22.27, EVID 2.95 vs 1.80 cm

Figure 4. Frames 0, 8, and 15 of the real video (top rows) and of IRASim-Frame-Ada from the same actions (bottom rows); both judges agree on each verdict. The EVID errors are given for the generated and the real video. In (b) the arm follows the actions and the PSNR is the highest of the three, yet the object vanishes, which neither metric detects; in (c) the contact failure also shows in the cycle error.

Limitations. *L1, objects do not keep their identity.* The most frequent violation found by the judges is an object that changes shape or vanishes, typically once the gripper holds it (Fig. 4b). *L2, correct motion does not imply the correct outcome.* Only 50% and 75% of Frame-Ada clips end in the same state as the real clip according to the two judges. *L3, errors grow with contact and time.* The gap between the cycle error of Frame-Ada and that of the real video is largest for wipe (Fig. 4c), turn, slide, and topple, at 0.5 to 0.7 cm, and close to zero for open, flip, and fold; within a clip the error grows towards the last frames. *L4, short clips of one embodiment.* Each checkpoint covers one dataset at 256×320 and 16 frames, about 3 s; longer horizons need autoregressive chaining, which we do not test.

8. Conclusion

On real robot data, frame-level appearance metrics cannot tell whether the robot in a generated video followed its actions: SSIM rates a static video as high as the generation from the correct actions, while trajectory-based scores degrade as the actions get worse. EVID provides this check without any real video, by reading the actions back with an off-the-shelf inverse dynamics model decoded exactly as its authors encode them: IRASim-Frame-Ada follows Bridge actions to within about 0.3 cm of what the same model reads from real videos, Cosmos-Predict2.5 with actions comes closer still, and the same backbone without actions moves the robot no better than a static frame. Almost every arm motion read from the generated videos is executable by the WidowX 250, so these models fail on objects, which change or vanish, rather than on the arm. Selecting the best of six IRASim samples by EVID did not improve independent scores, since the samples differ little. The study is limited to a few hundred clips, one embodiment for EVID, judges that are not stable between runs, and an inverse dynamics model whose own error of 1.50 cm bounds what EVID can resolve and whose training data may include Bridge test episodes. Replaying the motion read from a video in a physics simulator of the same robot, such as SIMPLER [7], would extend the executability check to dynamics, contacts, and collisions.

References

[1] Bowen Baker, Ilge Akkaya, Peter Zhokhov, Joost Huizinga, Jie Tang, Adrien Ecoffet, et al. Video PreTraining (VPT): Learning to act by watching unlabeled online videos. *arXiv preprint arXiv:2206.11795*, 2022.

[2] Hritik Bansal, Zongyu Lin, Tianyi Xie, Zeshun Zong, Michal Yarom, Yonatan Bitton, et al. VideoPhy: Evaluating physical commonsense for video generation. *arXiv preprint arXiv:2406.03520*, 2024.

[3] Yilun Du, Mengjiao Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. *arXiv preprint arXiv:2302.00111*, 2023.

[4] Gabriel Ilharco, Vihan Jain, Alexander Ku, Eugene Ie, and Jason Baldridge. General evaluation for instruction conditioned navigation using dynamic time warping. *arXiv preprint arXiv:1907.05446*, 2019.

[5] Joel Jang, Seonghyeon Ye, Zongyu Lin, Jiannan Xiang, Johan Bjorck, Yu Fang, et al. DreamGen: Unlocking generalization in robot learning through video world models. *arXiv preprint arXiv:2505.12705*, 2025.

[6] Nikita Karaev, Iurii Makarov, Jianyuan Wang, Natalia Neverova, Andrea Vedaldi, and Christian Rupprecht. CoTracker3: Simpler and better point tracking by pseudo-labelling real videos. *arXiv preprint arXiv:2410.11831*, 2024.

[7] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Karl Pertsch, Oier Mees, Homer Rich Walke, et al. Evaluating real-world robot manipulation policies in simulation. *arXiv preprint arXiv:2405.05941*, 2024.

[8] Corey Lynch, Ayzaan Wahid, Jonathan Tompson, Tianli Ding, James Betker, Robert Baruch, Travis Armstrong, and Pete Florence. Interactive language: Talking to robots in real time. *arXiv preprint arXiv:2210.06407*, 2022.

[9] Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? *arXiv preprint arXiv:2501.09038*, 2025.

[10] NVIDIA. Cosmos world foundation model platform for physical AI. *arXiv preprint arXiv:2501.03575*, 2025.

[11] NVIDIA. Cosmos3-Nano model card. https://huggingface.co/nvidia/Cosmos3-Nano, 2026. Accessed 2026-10-29.

[12] NVIDIA. Cosmos framework. https://github.com/NVIDIA/cosmos-framework, 2026. Commit 1a2f03a, accessed 2026-10-29.

[13] Maxime Oquab et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[14] William Peebles and Saining Xie. Scalable diffusion models with transformers. *arXiv preprint arXiv:2212.09748*, 2022.

[15] Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphael Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717*, 2018.

Source	Inputs	Outputs	Evaluation	Assumptions	Limitations
Video world models					
UniSim [18]	observation, language instruction or low-level control	video of the visual outcome	FID, FVD, IS, CLIP score in ablations; policies trained purely in the simulator and deployed zero-shot on real robots	mixing internet, simulation, and robot data yields one simulator of real interaction	hallucinated outcomes for unrealistic actions and limited memory, as stated by the authors
IRASim [20]	first frame, end-effector action trajectory	16 frames at 256×320	PSNR, SSIM, latent L2, FID, FVD against recorded videos; human preference; policy evaluation on LIBERO with human-judged success, Pearson 0.99 with the simulator; model-based planning	closeness to the recorded video measures simulation quality	action following is checked by frame metrics or by humans judging task success; one embodiment per checkpoint
Cosmos [10]	video and text; actions after post-training	video	3-D consistency by epipolar error and pose estimation; physics alignment by PSNR, SSIM, DreamSim, IoU on simulated scenes	pretraining on large video data transfers to physical AI after post-training	physics alignment measured on simulated scenes
Video models turned into actions by inverse dynamics					
UniPi [3]	first frame, text goal	video plan, then actions from an inverse dynamics model	task success in simulated manipulation; on Bridge, CLIP score, FID, FVD, and a success classifier on the last generated frame	a generated plan is executable and its actions can be recovered	slow diffusion sampling, as stated by the authors; plan errors reach the robot unchecked
DreamGen [5]	first frame, language, target robot	video, then pseudo-actions from a latent action or inverse dynamics model	DreamGen Bench with instruction following and physics alignment by VLM and human judges; downstream policy success	pseudo-actions from generated videos are useful training data	the benchmark scores the video, not the recovered actions
Evaluation of physical plausibility and of policies					
VideoPhy [2]	text prompt	text-to-video generation	human judgement of semantic adherence and physical commonsense; automatic judge VideoCon-Physics	physical commonsense can be judged from the video alone	no actions or robots; best model correct on 39.6% of prompts
Physics-IQ [9]	frames of a real physical experiment	predicted continuation	spatial and spatiotemporal IoU, weighted spatial IoU, and MSE of motion against the recorded continuation	the real continuation is the reference	needs the real video; no robot actions
SIMPLER [7]	robot policy	rollouts in simulated scenes matched to real setups	correlation of simulated and real success rates	reducing control and visual gaps suffices without full digital twins	evaluates policies rather than videos; few real setups

Table 5. Literature matrix. Evaluation methods are taken from each paper.

[16] Homer Walke, Kevin Black, Abraham Lee, Moo Jin Kim, Max Du, Chongyi Zheng, Tony Zhao, Philippe Hansen-Estruch, Quan Vuong, Andre He, Vivek Myers, Kuan Fang, Chelsea Finn, and Sergey Levine. BridgeData V2: A dataset for robot learning at scale. *arXiv preprint arXiv:2308.12952*, 2023.

[17] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[18] Sherry Yang, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Leslie Kaelbling, Dale Schuurmans, and Pieter Abbeel. Learning interactive real-world simulators. In *International Conference on Learning Representations*, 2024.

[19] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, 2018.

[20] Fangqi Zhu, Hongtao Wu, Song Guo, Yuxiao Liu, Chilam Cheang, and Tao Kong. IRASim: A fine-grained world model for robot manipulation. *arXiv preprint arXiv:2406.14540*, 2024.